Rabiul Awal

Department of Computer Science and Operations Research
University of Montreal, QC, Canada
Mila - Quebec AI Institute, QC, Canada

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Website: https://rabiul.me/
Research Interests: Generative Models, World Models, Reinforcement Learning

Current Position

09/2024 – present **Mila - Quebec AI Institute, Montreal**
Graduate Student Researcher

Education

09/2024 – present **University of Montreal**
Ph.D. in Computer Science
Department of Computer Science and Operations Research
Advisor: Aishwarya Agrawal
Courses: Reinforcement Learning and Optimal Control, Vision and Language
CGPA: 4.15/4.3

01/2021 – 12/2022 **University of Saskatchewan**
MSc. Computer Science
Advisor: Michael Horsch, Roy Ka-Wei Lee
CGPA: 88/100

03/2014 – 12/2018 **Noakhali Science and Technology University, Bangladesh**
BSc. Computer Science

Previous Experience

05/2026 – 09/2026 **Qualcomm Research, Amsterdam**
Research Intern

07/2024 – 09/2025 **ServiceNow Research, Montreal**
Visiting Researcher in the Multimodal Team
Advisor: Sai Rajeswar

01/2023 – 08/2024 **Mila - Quebec AI Institute, Montreal**
Research Intern
Advisor: Aishwarya Agrawal

06/2020 – 11/2022 **Singapore University of Technology and Design**
Student Researcher at Social AI Studio
Advisor: Roy Ka-Wei Lee

12/2019 – 12/2020 **Goava Sales Intelligence AB, Dhaka**
Software Engineer

In-Progress Research

[1] **Self-supervised Free Lunch for Flow Map Training**
An almost free feature-space consistency objective that accelerates training and improves the stability of one-/few-step diffusion models. [pdf]

[2] **Denoising Embedding Models are Self-Supervised Learners**
A unified denoising framework that serves as both a self-supervised visual encoder and an image generator, learning a generative embedding space. [pdf]

Peer-Reviewed Workshop Publications (Non-Archival)

[W2] **One Flow Transformer for Imagination and Control**
Rabiul Awal, Jinseong Jeong, Ankur Sikarwar, Parisa Kordjamshidi, Andrii Zadaianchuk, Sai Rajeswar, Paul Hongsuck Seo, Aishwarya Agrawal
RLC 2026 Workshop on Model-Based RL in the Era of Generative World Models

[W1] **Dynamics Forcing for Diffusion World Models**
Rabiul Awal
RLC 2026 Workshop on Model-Based RL in the Era of Generative World Models

Peer-Reviewed Conference Publications

Full list on Google Scholar. * denotes equal contribution.

[C8] **Grounding Computer Use Agents on Human Demonstrations**
Aarash Feizi, Shravan Nayak, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, **Rabiul Awal** + 11 others
ICLR 2026

[C7] **Rendering-Aware Reinforcement Learning for Vector Graphics Generation**
Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Aarash Feizi, Rishav Pramanik, Pascal Wichmann, Arnab Mondal, Mohammad Reza Samsami, **Rabiul Awal** + 7 others
NeurIPS 2025

[C6] **The Promise of RL for Autoregressive Image Editing**
Saba Ahmadi*, **Rabiul Awal***, Ankur Sikarwar*, Amirhossein Kazemnejad*, Ge Ya Luo, Juan A. Rodriguez, Sai Rajeswar, Siva Reddy, Christopher Pal, Benno Krojer, Aishwarya Agrawal
NeurIPS 2025

[C5] **CTRL-O: Language-Controllable Object-Centric Visual Representation Learning**
Aniket Didolkar*, Andrii Zadaianchuk*, **Rabiul Awal***, Maximilian Seitzer, Efstratios Gavves, Aishwarya Agrawal
CVPR 2025
Spotlight presentation at the Multimodal Algorithmic Reasoning Workshop, CVPR 2025

[C4] **WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation**
Rabiul Awal, Mahsa Massoud, Aarash Feizi, Zichao Li + 9 others
EMNLP 2025
Spotlight presentation at the Multimodal Algorithmic Reasoning Workshop, CVPR

[C3] **VisMin: Visual Minimal-Change Understanding**
Rabiul Awal*, Saba Ahmadi*, Le Zhang*, Aishwarya Agrawal
NeurIPS 2024
Contributed talk at the SpLU-RoboNLP Workshop, ACL 2024

[*C2*] **Benchmarking Vision Language Models for Cultural Understanding**
Shravan Nayak, Kanishk Jain, **Rabiul Awal**, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, Aishwarya Agrawal
EMNLP 2024
Oral presentation

[*C1*] **Contrasting Intra-modal and Ranking Cross-modal Hard Negatives to Enhance Visio-linguistic Compositional Understanding**
Le Zhang, **Rabiul Awal**, Aishwarya Agrawal
CVPR 2024
Spotlight presentation at the Open-Domain Reasoning Multi-Modal Workshop, CVPR 2023

Selected Awards, Grant & Fellowship

2026 Fonds de recherche du Québec – Nature et technologies (FRQNT) Scholarship ($25,000)

2026 DIRO Excellence Scholarship for the Winter 2026 session ($3,900)

2025 DIRO Excellence Scholarship for the Fall 2025 session ($3,000)

2025 Outstanding Reviewer at CVPR

2024 Selected for CIFAR DLRL Summer School at Toronto

2024 Selected for Upper Bound Talent Bursary at Alberta ($1,500)

2022 Selected for Alberta University AI Week Talent Bursary ($1,500)

Service

Broader Community

2024–present **Supervision Request Process Evaluation Committee**
Pre-evaluate MSc and PhD applications to help Mila professors recruit students

2025–present **Whiteboard Session Lead Organizer**
Organize the RL for Foundation Models reading group at Mila

2021–2022 **Computer Science Graduate Council MSc Representative**
Strengthened graduate research activities and hosted social events for graduate students at the University of Saskatchewan

2023–2024 **Language Grounding Reading Group Organizer**
Co-ran the vision-and-language reading group at Mila

Research Community

02/2026 **Workshop Organizer at Mila**
World Modeling Workshop

05/2025 **Workshop Organizer at CVPR 2025**
Vision Language Models For All: Building Geo-Diverse and Culturally Aware Vision-Language Models

2021–present **Reviewer**
AAAI26, NeurIPS25, CVPR25, ECCV24, CVPR24, ACL23, SIGIR22, AACL22, ICDM21, SIGKDD21